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GATE ECE · ANALOG CIRCUITS · CHAPTER-WISE PYQS

# Bipolar Junction Transistor

GATE ECE Analog Circuits · Previous Year Questions with Answer Key

| 78 | 39 | 1988 – 2026 | English |
|---|---|---|---|
| QUESTIONS | PAPERS COVERED | YEARS | LANGUAGE |

## INSIDE THIS PDF

| | |
|---|---|
| Marks 1 | 26 questions |
| Marks 2 | 46 questions |
| Marks 5 | 6 questions |
| Answer key | at the end |

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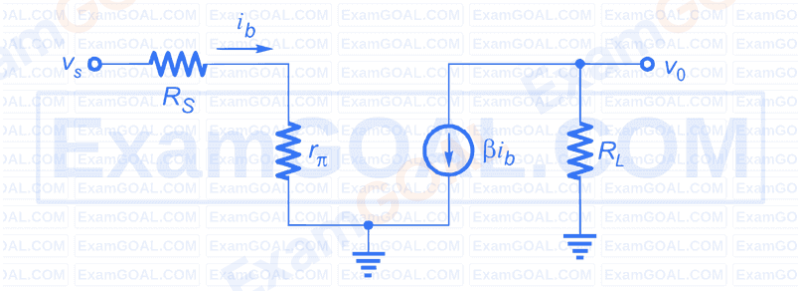
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2. **Check**
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3. **Understand**
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## Marks 1 26 questions

**Q1.** GATE ECE 2025

A simplified small-signal equivalent circuit of a BJT-based amplifier is given below. The small-signal voltage gain $V_o/V_s$ (in $\mathrm{V/V}$ ) is_____.

Open solution

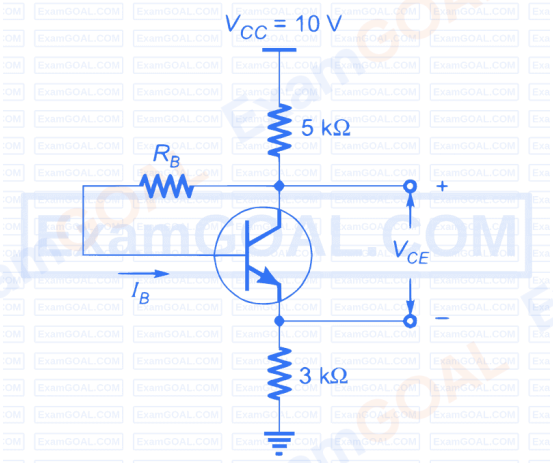



(A) $\frac{-\beta R_L}{R_s+r_\pi}$ (B) $\frac{+\beta R_L}{R_s}$ (C) $\frac{-\beta R_L}{R_s}$ (D) $\frac{+\beta R_L}{R_s+r_\pi}$

**Q2.** GATE ECE 2025

The ideal BJT in the circuit given below is biased in the active region with a $\beta$ of 100 . If $I_B$ is $10\mu$ A, then $V_{CE}$ (in Volts, rounded off to two decimal places) is _____ .

Open solution



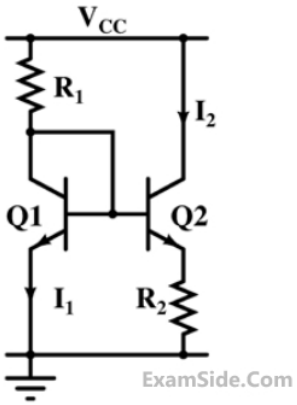

(A) 4.95 (B) 3.03 (C) 1.92 (D) 3.73

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**Q3.** GATE ECE 2016 Set 2

Open solution

Resistor $R_1$ in the circuit below has been adjusted so that $I_1$ = 1 mA. The bipolar transistor Q1 and Q2 are perfectly matched and have very high current gain, so their base currents are negligible. The supply voltage $V_{cc}$ is 6 V. The thermal voltage kT/q is 26 mV.



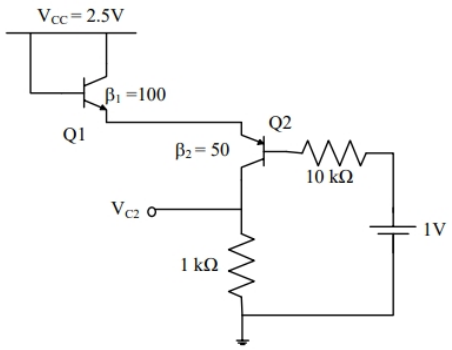
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The value of $R_2$ (in $\Omega$ ) for which $I_2$ = 100 $\mu$A is _____.

**Q4.** GATE ECE 2016 Set 3

Open solution

Consider the circuit shown in the figure. Assuming $V_{BE1} = V_{EB2}$ = 0.7 volt, value of the dc voltage $V_{c2}$ (in volt) is _____



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**Q5.** GATE ECE 2014 Set 4

Open solution

If the emitter resistance in a common-emitter voltage amplifier is not bypassed, it will

(A) reduce both the voltage gain and the input impedance

(B) reduce the voltage gain and increase the input impedance

(C) increase the voltage gain and reduce the input impedance

(D) increase both the voltage gain and the input impedance

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**Q6.** GATE ECE 2014 Set 2

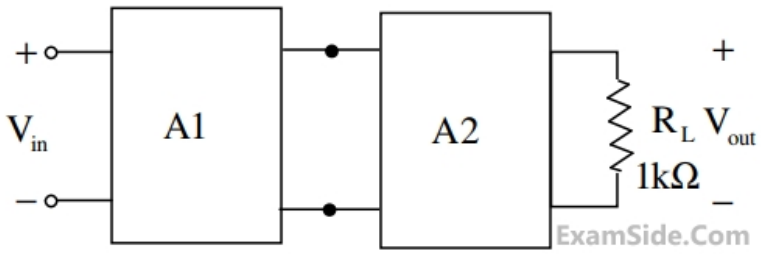
Open solution

A cascade connection of two voltage amplifiers A1 and A2 is shown in the figure. The open-loop gain $A_{v0}$, input resistance $R_{in}$, and output resistance $R_0$ for A1 and A2 are as follows:

A1: $A_{v0}$ = 10,$R_{in}$ = 10 kΩ ,$R_0$ =1 kΩ

A2 : $A_{v0}$ = 5,$R_{in}$ = 5 kΩ , $R_0$ = 200 Ω

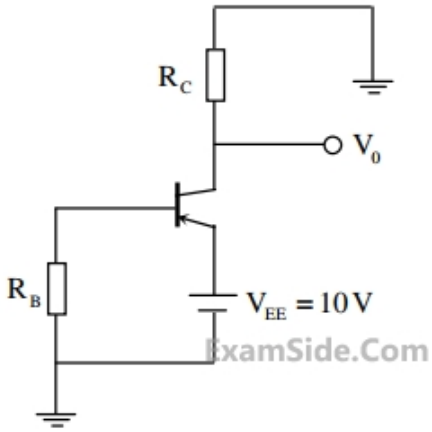
The approximate overall voltage gain $\frac{V_{out}}{V_{in}}$ is _____.




**Q7.** GATE ECE 2014 Set 3

Open solution

In the circuit shown, the PNP transistor has $|V_{BE}| = 0.7$ V and β = 50. Assume that $R_B$ = 100 kΩ. For $V_0$ to be 5 V, the value of $R_C$ ( in kΩ) is _____.

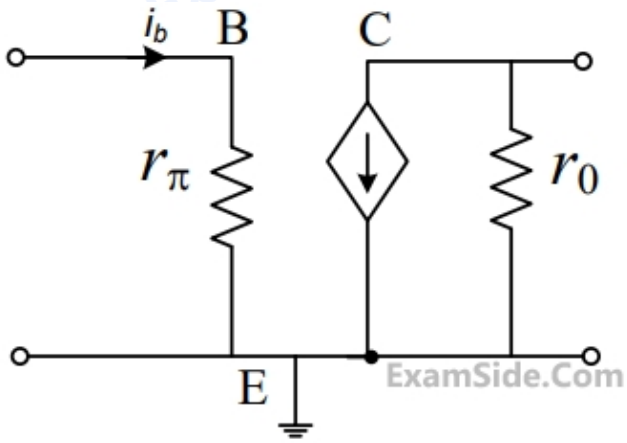



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**Q8.** GATE ECE 2012

Open solution

The current $i_b$ through the base of a silicon npn transistor is $1 + 0.1 \cos(10000 \pi t)$ mA . At 300 K, the $r_\pi$ in the small signal model of the transistor is

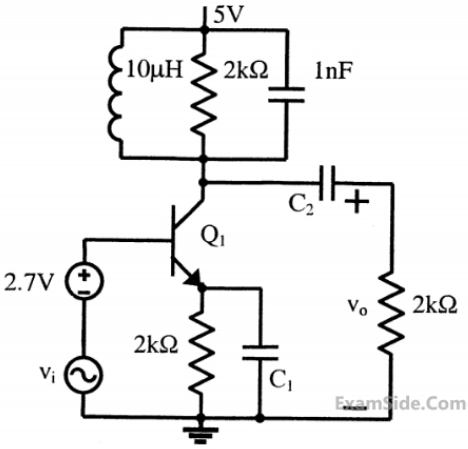



(A) $250\ \Omega$ (B) $27.5\ \Omega$ (C) $25\ \Omega$ (D) $22.5\ \Omega$

**Q9.** GATE ECE 2011

Open solution

In the circuit shown below, capacitors $C_1$ and $C_2$ are very large and are shorts at the input frequency. $V_i$ is a small signal input. The gain magnitude $\left|\frac{V_0}{V_i}\right|$ at 10 Mrad/s is



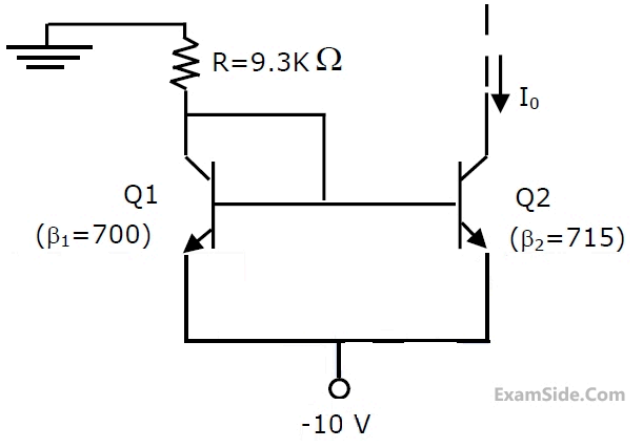

(A) maximum (B) minimum (C) unity (D) zero

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**Q10.** GATE ECE 2010

Open solution

In the silicon BJT circuit shown below, assume that the emitter area of transistor $Q_1$ is half that of transistor $Q_2$.The value of current $I_0$ is approximately

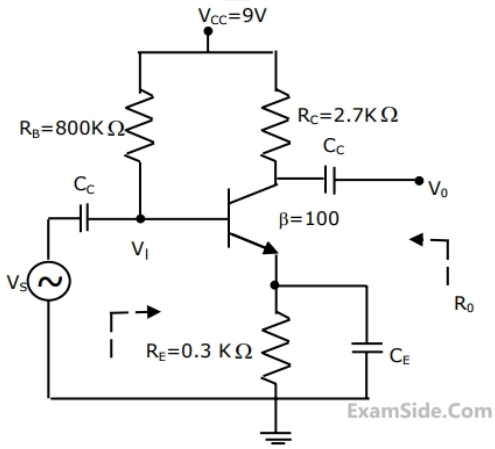



(A) 0.5 mA (B) 2 mA (C) 9.3 mA (D) 15 mA

**Q11.** GATE ECE 2010

Open solution

The amplifier circuit shown below uses a silicon transistor. The capacitors $C_C$ and $C_E$ can be assumed to be short at signal frequency and the effect of output resistance $r_0$ can be ignored. If $C_E$ is disconnected from the circuit, which one of the following statements is TRUE?




(A) The input resistance $R_i$ increases and the magnitude of voltage gain $A_V$ decreases

(B) The input resistance $R_i$ decreases and the magnitude of voltage gain $A_V$ decreases

(C) Both input resistance $R_i$ and the magnitude of voltage gain $A_V$ decrease

(D) Both input resistance $R_i$ and the magnitude of voltage gain $A_V$ increase

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**Q12.** GATE ECE 2006

The input impedance ($Z_i$) and the output impedance ($Z_o$) of an ideal transconductance (voltage controlled current source) amplifier are

(A) $Z_i = 0, Z_o = 0$ (B) $Z_i = 0, Z_o = \infty$

(C) $Z_i = \infty, Z_o = 0$ (D) $Z_i = \infty, Z_o = \infty$

Open solution

**Q13.** GATE ECE 2005

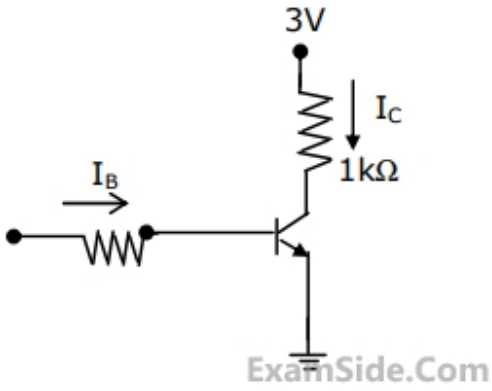
The cascode amplifier is a multistage configuration of

(A) CC-CB (B) CE-CB (C) CB-CC (D) CE-CC

Open solution

**Q14.** GATE ECE 2004

Assuming $V_{Cesat}$=0.2V and β = 50, the minimum base current ($I_B$) required to drive the transistor in the figure to saturation is



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(A) 56 μA (B) 140 mA (C) 60 μA (D) 3 mA

Open solution

**Q15.** GATE ECE 2001

The current gain of a BJT is

(A) $g_m r_o$ (B) $\frac{g_m}{r_o}$ (C) $g_m r_\pi$ (D) $\frac{g_m}{r_\pi}$

Open solution

**Q16.** GATE ECE 2000

Introducing a resistor in the emitter of a common amplifier, stabilizes the dc operating point against variations in

(A) only the temperature (B) only the β of the transistor

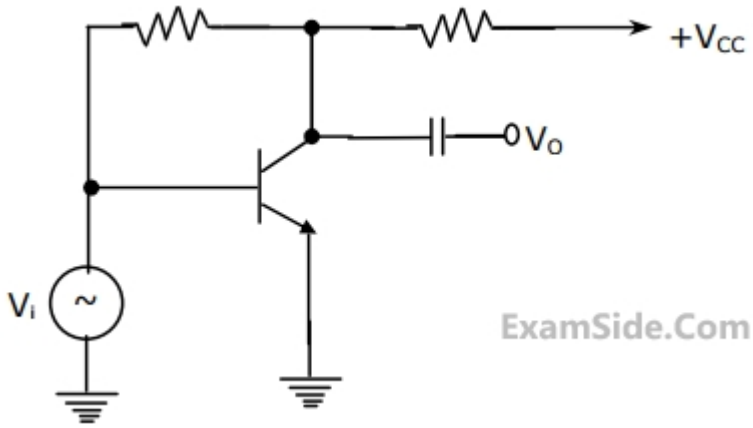
(C) both temperature and β (D) none of the above

Open solution

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**Q17.** GATE ECE 1998

The circuit of Fig. is an example of feedback of the following type




(A) current series (B) current shunt (C) voltage series (D) voltage shunt

Open solution

**Q18.** GATE ECE 1998

From a measurement of the rise time of the output pulse of an amplifier whose input is a small amplitude square wave, one can estimate the following parameter of the amplifier:

(A) gain-bandwidth product (B) slow rate

(C) upper 3-dB frequency (D) lower 3-dB frequency

Open solution

**Q19.** GATE ECE 1998

The emitter coupled pair of BJT's gives a linear transfer relation between the differential output voltage and the differential input voltage $V_{id}$ only. When the magnitude of $V_{id}$ is less than '$\alpha$' times the thermal voltage, where '$\alpha$' is

(A) 4 (B) 3 (C) 2 (D) 1

Open solution

**Q20.** GATE ECE 1998

In a series regulated power supply circuit, the voltage gain $A_v$ of the 'pass' transistor satisfies the condition:

(A) $A_v \rightarrow \infty$ (B) $1 \ll A_v \ll \infty$ (C) $A_v \approx 1$ (D) $A_v \ll 1$

Open solution

**Q21.** GATE ECE 1995

A BJT is said to be operating in the saturation region if

(A) both the junctions are reverse biased

(B) base-emitter junction is reverse biased and base-collector junction is forward biased.

(C) base-emitter junction is forward biased and base-collector junction reverse biased

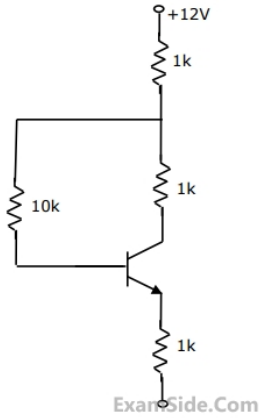
(D) both the junctions are forward biased

Open solution

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**Q22.** GATE ECE 1995

A transistor having $\alpha$ = 0.99 and $V_{BE}$ = 0.7 V, is used in the circuit of the figure.What is the value of the collector current?




Open solution

(A) 5.32 mA (B) 6.0 mA (C) 6.5 mA (D) 7.0 mA

**Q23.** GATE ECE 1994

In order to reduce the harmonic distortion in an amplifier, its dynamic range has to be _____

Open solution

**Q24.** GATE ECE 1994

A common emitter transistor amplifier has a collector current of 1.0 mA when its base current is 25 $\mu$A at the room temperature. It's input resistance is approximately equal to (in K$\Omega$) _____ .

Open solution

**Q25.** GATE ECE 1993

The bandwidth of an n-stage tuned amplifier, with each stage having a band width of B, is given by

(A) B/n (B) $B/\sqrt{n}$

(C) $B\sqrt{2^{\frac{1}{n}}-1}$ (D) $B/\sqrt{2^{\frac{1}{n}}-1}$

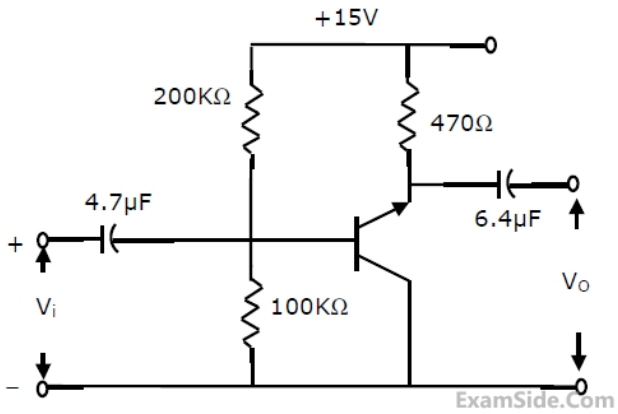
Open solution

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**Q26.** GATE ECE 1993

Open solution

For the amplifier circuit of figure, the transistor has a $\beta$ of 800. The mid-band voltage gain $V_0/V_1$ of the circuit will be




(A) $0$ (B) < 1 (C) $\approx 1$ (D) 800

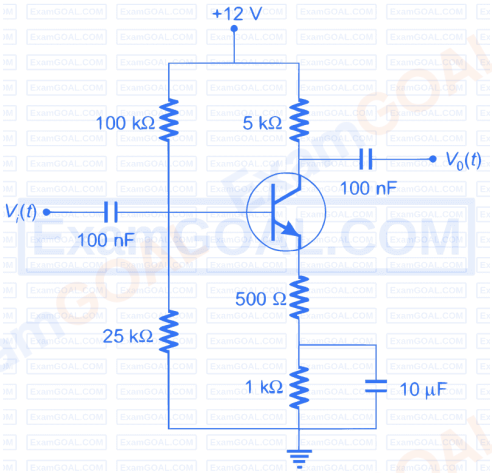
## Marks 2 46 questions

**Q27.** GATE ECE 2026

Open solution

A small signal source, $V_i(t) = A\cos\left(10^5 t\right) + B\sin\left(10^7 t\right)$ is applied to a BJT circuit as shown in the Figure.

Assume zero source resistance, $V_{BE} = 0.7\ \text{V}, \beta_{\text{dc}} = 99$, Early voltage $= 100\ \text{V}$ and Thermal voltage $= 25\text{mV}$. Effect of internal parasitic capacitances of the BJT may be neglected. Which expression is the best approximation of the output voltage $V_o(t)$ ?




(A) $-9.1\left[A\cos\left(10^5 t\right) + B\sin\left(10^7 t\right)\right]$

(B) $9.1\left[A\cos\left(10^5 t\right) + B\sin\left(10^7 t\right)\right]$

(C) $-190.4\left[A\cos\left(10^5 t\right) + B\sin\left(10^7 t\right)\right]$

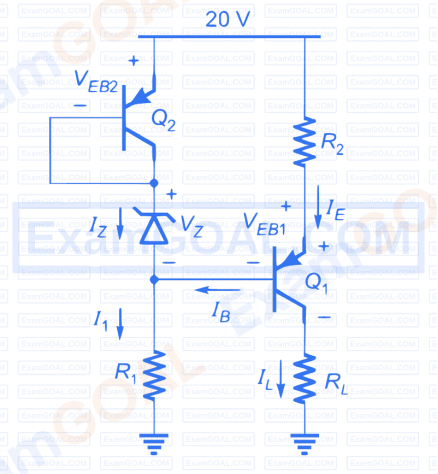
(D) $190.4\left[A\cos\left(10^5 t\right) - B\sin\left(10^7 t\right)\right]$

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**Q28.** GATE ECE 2025

Open solution

In the circuit shown, the identical transistors Q1 and Q2 are biased in the active region with $\beta = 120$. The Zener diode is in the breakdown region with $V_Z = 5\ \text{V}$ and $I_Z = 25\ \text{mA}$. If $I_L = 12\ \text{mA}$ and $V_{EB1} = V_{EB2} = 0.7\ \text{V}$, then the values of $R_1$ and $R_2$ (in $\text{k}\Omega$, rounded off to one decimal place) are _____, respectively.




(A) 0.6 and 0.4 (B) 1.4 and 2.5 (C) 14.0 and 25.0 (D) 6.0 and 4.0

**Q29.** GATE ECE 2024

Open solution

Which of the following statements is/are true for a BJT with respect to its DC current gain $\beta$?

(A) Under high-level injection condition in forward active mode, $\beta$ will decrease with increase in the magnitude of collector current.

(B) Under low-level injection condition in forward active mode, where the current at the emitter-base junction is dominated by recombination-generation process, $\beta$ will decrease with increase in the magnitude of collector current.

(C) $\beta$ will be lower when the BJT is in saturation region compared to when it is in active region.

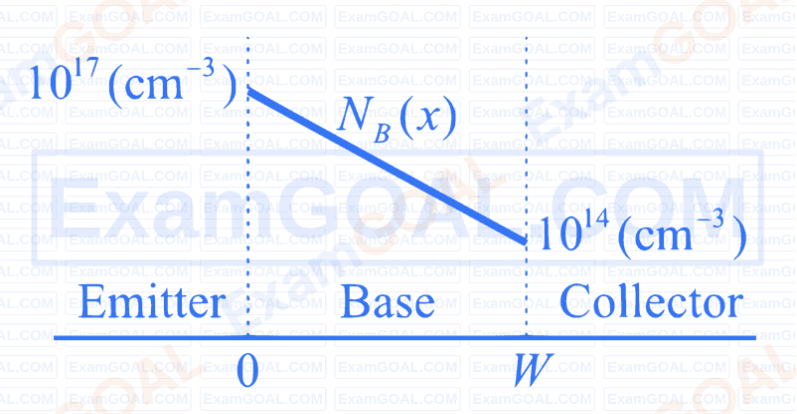
(D) A higher value of $\beta$ will lead to a lower value of the collector-to-emitter breakdown voltage.

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**Q30.** GATE ECE 2020

Open solution

The base of an $npn$ BJT $T1$ has a linear doping profile $N_B(x)$ as shown below. The base of another $npn$ BJT $T2$ has a uniform doping $N_B$ of $10^{17}\ \text{cm}^{-3}$. All other parameters are identical for both the devices. Assuming that the hole density profile is the same as that of doping, the common - emitter current gain of $T2$ is




(A) approximately 2.5 times that of $T1$.

(B) approximately 2.0 times that of $T1$.

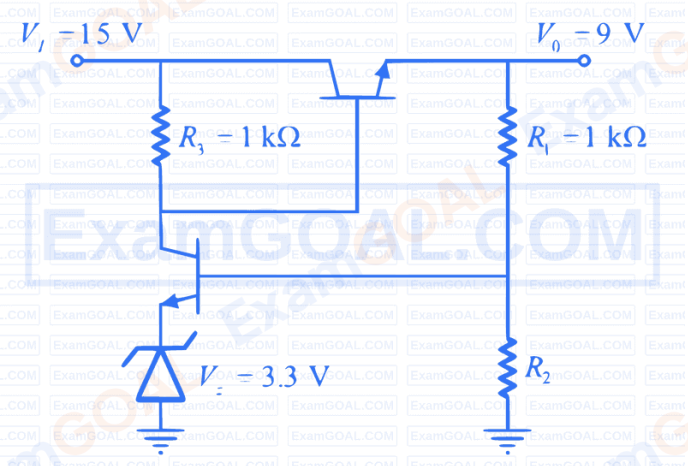
(C) approximately 3.0 times that of $T1$.

(D) approximately 0.7 times that of $T1$.

**Q31.** GATE ECE 2020

Open solution

In the voltage regulator shown below, $V_I$ is the unregulated input at 15 V . Assume $V_{BE} = 0.7$ V and the base current is negligible for both the BJTs. If the regulated output $V_0$ is 9 V , the value of $R_2$ is ____ $\Omega$.

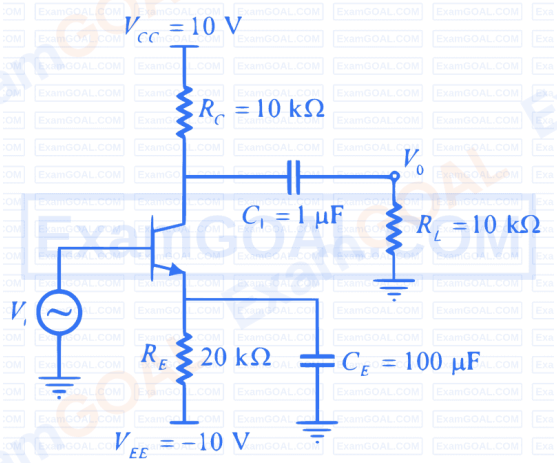



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**Q32.** GATE ECE 2020

Open solution

For the BJT in the amplifier shown below, $V_{BE} = 0.7\ \text{V}$, $\frac{kT}{q} = 26\text{mV}$. Assume that the BJT output resistance ( $r_0$ ) is very high and the base current is negligible. The capacitors are also assumed to be short circuited at signal frequencies. The input $V_i$ is direct coupled. The low frequency voltage gain $\frac{V_0}{V_i}$ of the amplifier is

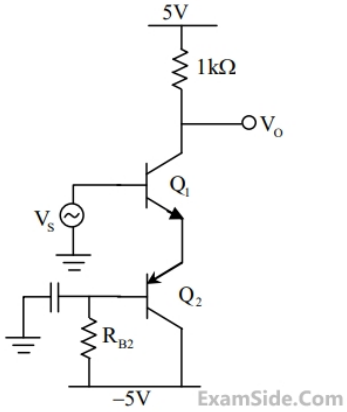



(A) -256.42 (B) -128.21 (C) -89.42 (D) -178.85

**Q33.** GATE ECE 2017 Set 2

Open solution

In the circuit shown, transistors $Q_1$ and $Q_2$ are biased at a collector current of 2.6 mA. Assuming that transistor current gains are sufficiently large to assume collector current equal to emitter current and thermal voltage of 26m V, the magnitude of voltage gain $V_o/V_s$ in the mid-band frequency range is _____ (up to second decimal place).

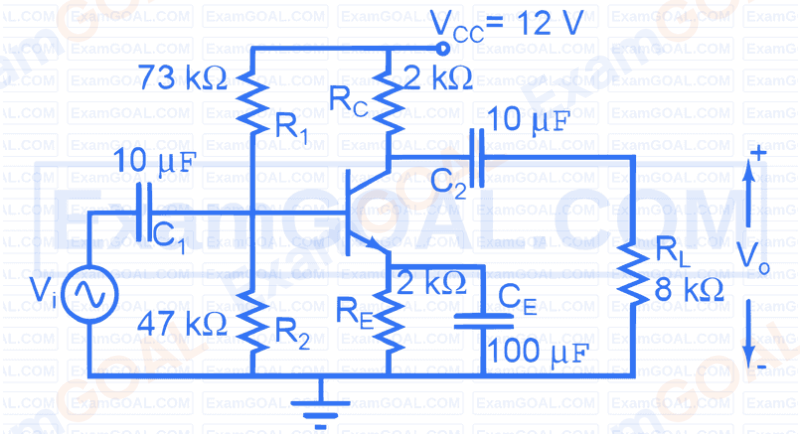



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**Q34.** GATE ECE 2017 Set 1

Open solution

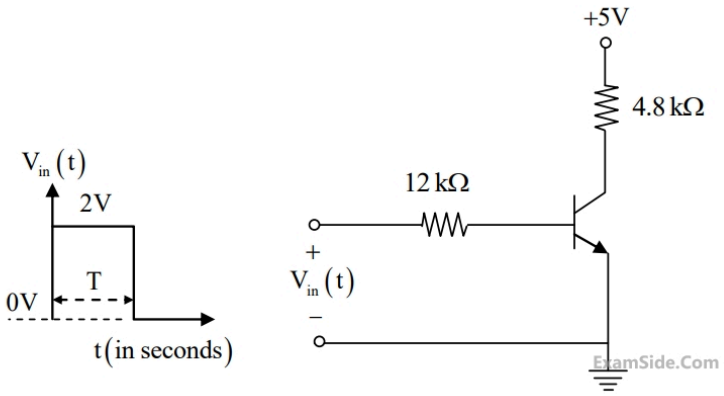
For the DC analysis of the Common-Emitter amplifier shown, neglect the base current and assume that the emitter and collector currents are equal. Given that $V_T$ = 25mV, $V_{BE}$ = 0.7V, and the BJT output resistance $r_0$ is practically infinite. Under these conditions the midband voltage gain magnitude, $a_v = |v_0/v_i|$ $V/V$, is _____




**Q35.** GATE ECE 2017 Set 1

Open solution

In the figure shown, the npn transistor acts as a switch.



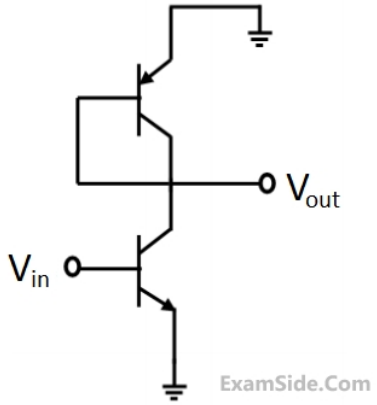

For the input $v_{in}(t)$as shown in the figure, the transistor switches between the cut-off and saturation regions of operation, when T is large. Assume collector-to-emitter voltage saturation $V_{CE}$(sat) = 0.2V and base-to-emitter voltage $V_{BE}$ = 0.7V. The minimum value of the common-base current gain$(\alpha)$ of the transistor for the switching should be _____.

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**Q36.** GATE ECE 2015 Set 2

In the ac equivalent circuit shown, the two **BJTs** are biased in active region and have identical parameters with β >> 1. The open circuit small signal voltage gain is approximately _____.

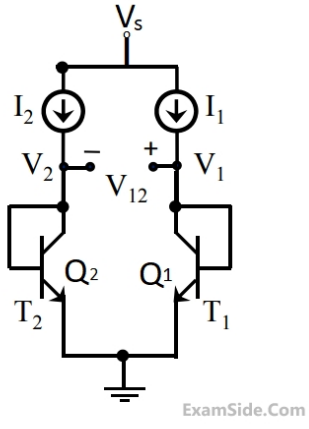
Open solution




**Q37.** GATE ECE 2015 Set 1

In the circuit shown, $I_1$ = 80 mA and $I_2$ b= 4mA. Transistors $T_1$ and $T_2$ are identical. Assume that the thermal voltage $V_T$ is 26 mV at 27°C. At 50°C, the value of the voltage $V_{12} = V_1 - V_2$ (in mV) is _____.

Open solution




**Q38.** GATE ECE 2014 Set 1

A BJT is biased in forward active mode. Assume $V_{BE}$ = 0.7 V, kT/q = 25 mV and reverse saturation current $I_s = 10^{-13}$A. The transconductance of the BJT (in mA/V) is

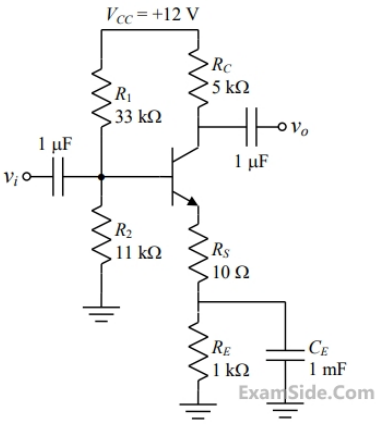
Open solution

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**Q39.** GATE ECE 2014 Set 1

For the amplifier shown in the figure, the BJT parameters are $V_{BE}$ = 0.7 V, $\beta$ = 200, and thermal voltage $V_T$ = 25 mV. The voltage gain ($V_o/V_i$ of the amplifier is _____.

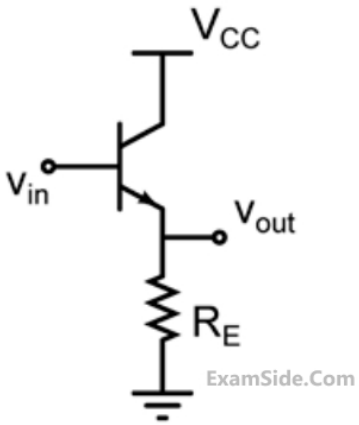
Open solution




**Q40.** GATE ECE 2014 Set 4

Consider the common-collector amplifier in the figure (bias circuitry ensures that the transistor operates in forward active region, but has been omitted for simplicity). Let $I_C$ be the collector current, $V_{BE}$ be the base-emitter voltage and $V_T$ be the thermal voltage. Also, $g_m$ and $r_o$ are the small-signal transconductance and output resistance of the transistor, respectively. Which one of the following conditions ensures a nearly constant small signal voltage gain for a wide range of values of $R_E$?

Open solution




(A) $g_m R_E << 1$ (B) $I_C R_E >> V_T$ (C) $g_m R_o >> 1$ (D) $V_{BE} >> V_T$

**Q41.** GATE ECE 2014 Set 4

A BJT in common-base configuration is used to amplify a signal received by a $50\,\Omega$ antena. Assume kT/q = 25 mV. The value of the collector bias current ( in mA ) required to match the input impedance of the amplifier to the impedance of the antena is _____.

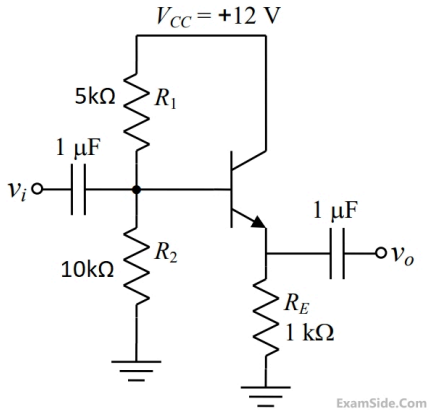
Open solution

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**Q42.** GATE ECE 2014 Set 4

Open solution

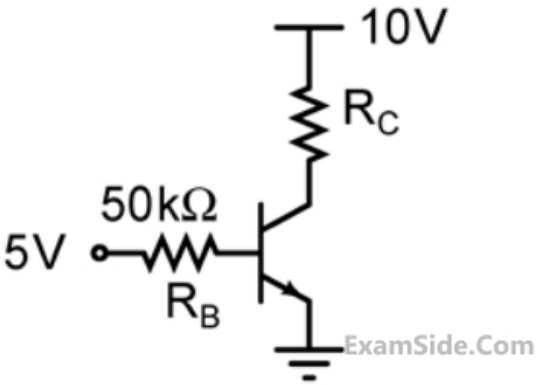
For the common collector amplifier shown in the figure, the BJT has high $V_{CE(sat)}$, and $V_{BE} = 0.7\ V$. The maximum undistorted peak-to-peak output voltage $V_o$ (in Volts) is ____.




**Q43.** GATE ECE 2014 Set 3

Open solution

In the circuit shown, the silicon BJT has $\beta\ \ 50.$ Assume $V_{BE} = 0.7$ and $V_{CE(sat)} = 0.2V$. Which one of the following statements is correct ?




(A) For $R_c = 1k\Omega$, the BJT operates in the saturation region.

(B) For For $R_c = 3k\Omega$, the BJT operates in the saturation region

(C) For For $R_c = 20k\Omega$, the BJT operates in the cut-off region.

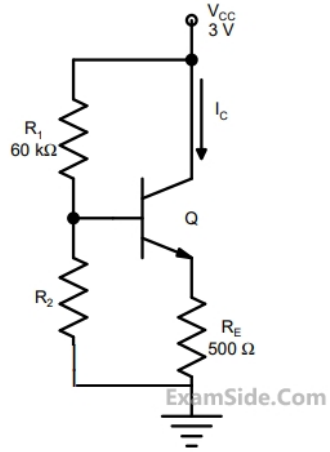
(D) For For $R_c = 20k\Omega$, the BJT operates in the linear region.

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**Q44.** GATE ECE 2013

Open solution

In the circuit shown below, the silicon npn transistor Q has a very high value of $\beta$. The required value of $R_2$ in k$\Omega$ to produce $I_C$ = 1mA is

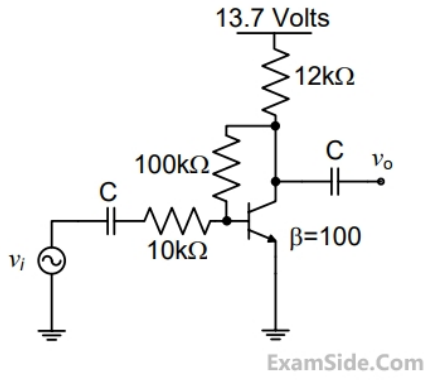



(A) 20 (B) 3 (C) 40 (D) 50

**Q45.** GATE ECE 2012

Open solution

The voltage gain $A_V$ of the circuit shown below is




(A) $|A_v| \approx 200$ (B) $|A_v| \approx 100$
(C) $|A_v| \approx 20$ (D) $|A_v| \approx 10$

**Q46.** GATE ECE 2011

Open solution

For a BJT, the common-base current gain $\alpha = 0.98$ and the colector base junction reverse bias saturation current $I_{CO}$ = 0.6 $\mu$A. This BJT is connected in the common emitter mode and operated in the active region with a base drive current $I_B$ = 20$\mu$A. The collector current $I_C$ for this mode of operation is

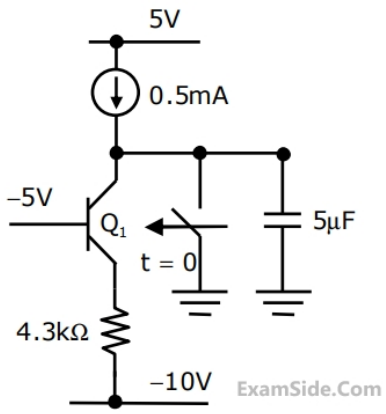
(A) 0.98 mA (B) 0.99 mA (C) 1.0 mA (D) 1.01 mA

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**Q47.** GATE ECE 2011

Open solution

For the BJT $Q_1$ in the circuit shown below, $\beta = \infty$, $V_{BEon} = 0.7V$, $V_{CEsat} = 0.7V$.. The switch is initially closed. at time t = 0, the switch is opened. The time t at which $q_1$ leaves the active region is

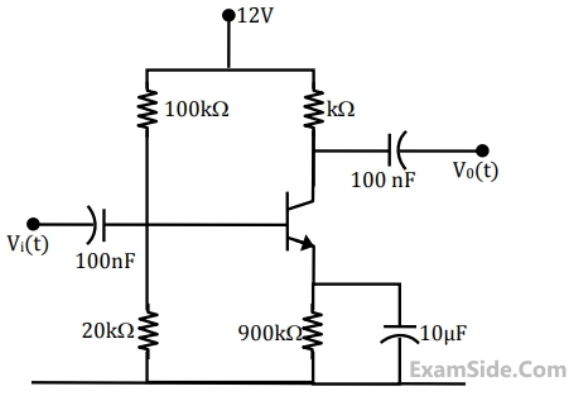



(A) 10 ms (B) 25 ms (C) 50 ms (D) 100 ms

**Q48.** GATE ECE 2009

Open solution

A small signal source $V_i(t)$ = A cos 20t + B sin $10^6$ t, is applied to a transistor Amplifier as shown in fig. The transistor has $\beta$ = 150 and $h_{ie}$ = 3kΩ , which expression best approximates $V_0(t)$ ?




(A) $V_o(t) = -1500\ (\mathrm{A}\cos 20t + B\sin 10^6 t)$

(B) $V_o(t) = -150\ (\mathrm{A}\cos 20t + B\sin 10^6 t)$

(C) $V_o(t) = -1500\, B\sin 10^6 t$

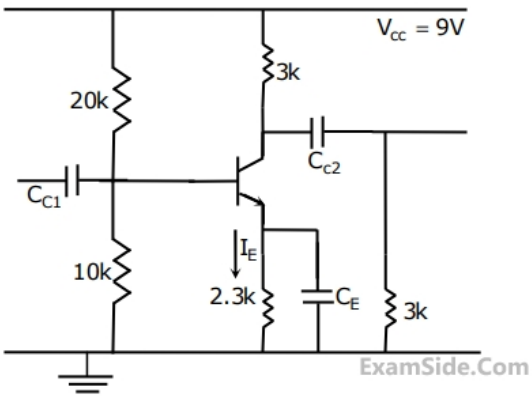
(D) $V_o(t) = -150\, B\sin 10^6 t$

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**Q49.** GATE ECE 2008

Open solution

In the following transistor circuit, $V_{BE}$ = 0.7V, $r_e$ = 25m/$I_E$, $\beta$ and all the capacitances are very large.



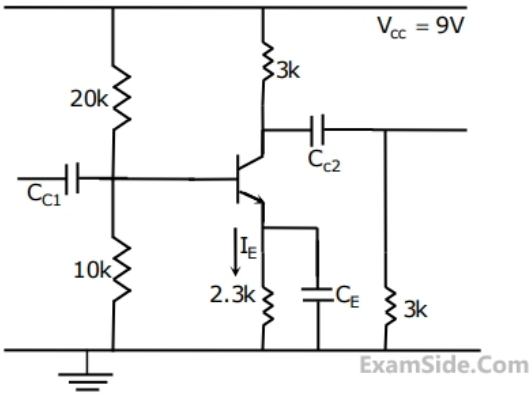

The Value of DC current $I_E$ is

(A) 1 mA (B) 2 mA (C) 5 mA (D) 10 mA

**Q50.** GATE ECE 2008

Open solution

In the following transistor circuit, $V_{BE}$ = 0.7V, $r_e$ = 25m/$I_E$, $\beta$ and all the capacitances are very large.




The mid-band voltage gain of the amplifier is approximately

(A) -180 (B) -120 (C) -90 (D) -60

**Q51.** GATE ECE 2007

Open solution

In a transconductance amplifier , it is desirable to have

(A) a large input resistance and large output resistance

(B) a large input resistance and a small output resistance

(C) a small input resistance and a large output resistance

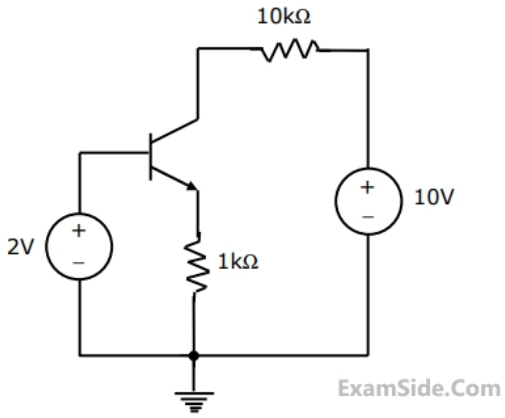
(D) a small input resistance and a small output resistance

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**Q52.** GATE ECE 2007

Open solution

For the BJT circuit shown, assume that the $\beta$ of the transistor is very large and $V_{BE}$ = 0.7 V. The mode of operation of the BJT is

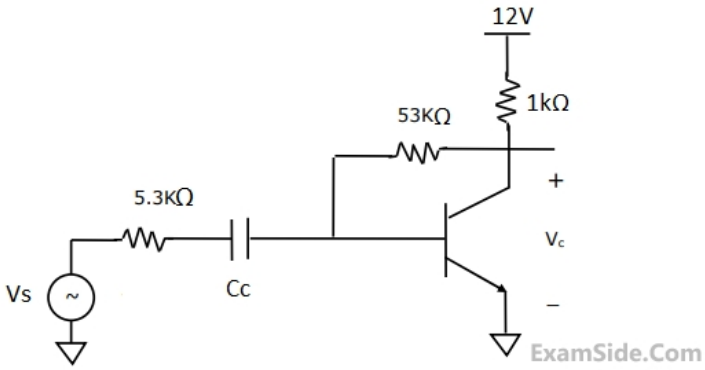



(A) cut-off (B) saturation (C) normal active (D) reverse active

**Q53.** GATE ECE 2006

Open solution

In the transistor amplifier circuit shown in the figure below, the transistor has the following parameters: $\beta_{DC}$ = 60, $V_{BE}$ = 0.7V, $h_{ie} \rightarrow \infty$, $h_{fe} \rightarrow \infty$. The capacitance $C_C$ can be assumed to be infinite.




Under the **DC** conditions, the collector-to emitter voltage drop is

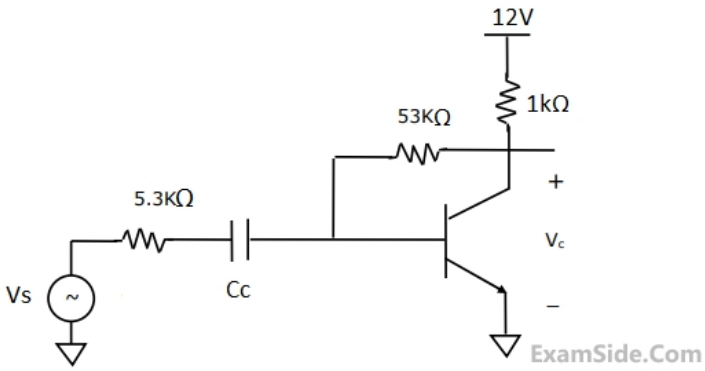
(A) 4.8 Volts (B) 5.3 Volts (C) 6.0 Volts (D) 6.6 Volts

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**Q54.** GATE ECE 2006

Open solution

In the transistor amplifier circuit shown in the figure below, the transistor has the following parameters: $\beta_{DC}$ = 60, $V_{BE}$ = 0.7V, $h_{ie} \rightarrow \infty$, $h_{fe} \rightarrow \infty$. The capacitance $C_C$ can be assumed to be infinite.



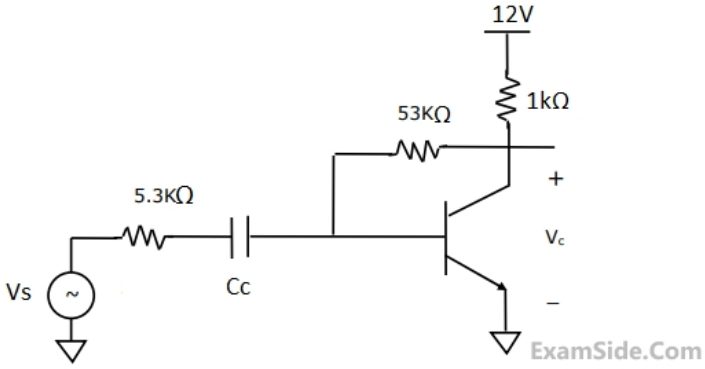

If $\beta_{DC}$ is increased by 10%, the collector-to-emitter voltage drop

(A) increase by less than (or) equal to 10%
(B) decreases by less than (or) equal to 10%
(C) increases by more than 10%
(D) decreases by more than 10%

**Q55.** GATE ECE 2006

Open solution

In the transistor amplifier circuit shown in the figure below, the transistor has the following parameters: $\beta_{DC}$ = 60, $V_{BE}$ = 0.7V, $h_{ie} \rightarrow \infty$, $h_{fe} \rightarrow \infty$. The capacitance $C_C$ can be assumed to be infinite.




The small-signal gain of the amplifier $\frac{V_c}{V_s}$ is

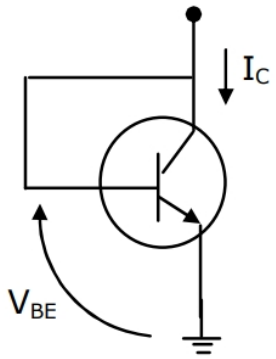
(A) -10
(B) -5.3
(C) 5.3
(D) 10

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**Q56.** GATE ECE 2005

Open solution

For an npn transistor connected as shown in the figure, $V_{BE}$ = 0.7 volts. Given that reverse saturation current of the junction at room temperature $300^0$ K is $10^{-13}$ A, the emitter current is $(\eta = 1)$



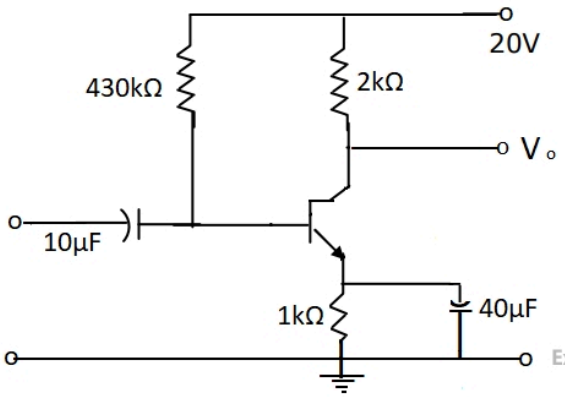

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(A) 30 mA (B) 39 mA (C) 49 mA (D) 20 mA

**Q57.** GATE ECE 2005

Open solution

The circuit using a **BJT** with β = 50 and $V_{BE}$ = 0.7 V is shown in the figure. The base current $I_B$ and collector voltage $V_C$ are respectively




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(A) 43 μA and 11.4 Volts (B) 40 μA and 16 Volts
(C) 45 μA and 11 Volts (D) 50 μA and 10 Volts

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**Q58.** GATE ECE 2004

A bipolar transistor is operating in the active region with a collector current of 1mA. Assuming that the 'β' of the transistor is 100 and the transconductance ($g_m$) and the input resistance ($r_\pi$) of the transistor in the common emitter configuration, are

Open solution

(A) $g_m$ = 25 mA/V and $r_\pi = 15.6\ k\Omega$

(B) $g_m$ = 40 mA/V and $r_\pi = 4.0\ k\Omega$

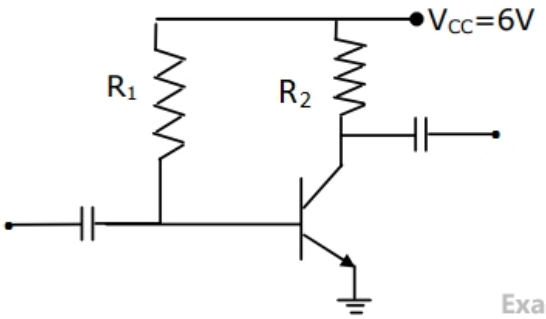
(C) $g_m$ = 25 mA/V and $r_\pi = 2.5\ k\Omega$

(D) $g_m$ = 40 mA/V and $r_\pi = 2.5\ k\Omega$

**Q59.** GATE ECE 2003

In the amplifier circuit shown in the figure, the values of $R_1$ and $R_2$ are such that the transistor is operating at $V_{CE}$ = 3V and $I_C$ = 1.5 mA when its $\beta$ is 150. For a transistor with $\beta$ of 200, the operating point ($V_{CE}$, $I_C$) is

Open solution




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(A) ( 2 V , 2 mA)

(B) ( 3 V , 2 mA)

(C) ( 4 V , 2 mA)

(D) ( 2 V , 1 mA)

**Q60.** GATE ECE 2003

Three identical amplifiers with each one having a voltage gain of 50, input Resistance of 1 KΩ and output resistance of 250Ω, are cascaded. The open circuit voltage gain of the combined Amplifier is

Open solution

(A) 49 dB

(B) 51 dB

(C) 98 dB

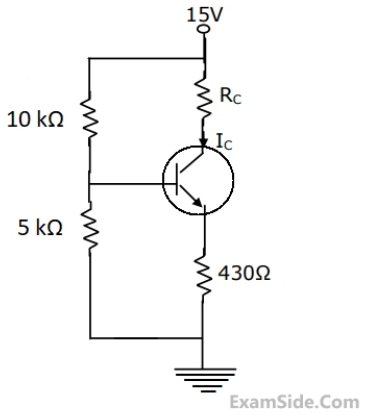
(D) 102 dB

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**Q61.** GATE ECE 2000

Open solution

In the circuit shown in the figure assume that the transistor is in the active region. It has a large β and its base Emitter voltage is 0.74. The value of $I_c$ is

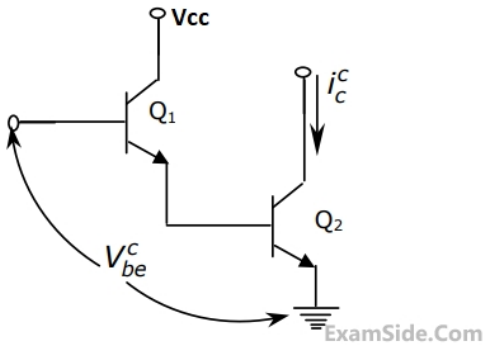



(A) indeterminate since $R_c$ is not given.
(B) 1 mA
(C) 5 mA
(D) 10 mA

**Q62.** GATE ECE 1996

Open solution

A Darlington stage is shown in the figure, if the transconductance of $q_1$ is $g_{m1}$ and $q_2$ is $g_{m2}$, then the overall transconductance $g_m^c \left[ \triangleq \frac{i_c^c}{v_{be}^c} \right]$ is given by




(A) $g_{m1}$
(B) 0.5 $g_{m1}$
(C) $g_{m2}$
(D) 0.5 $g_{m2}$

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**Q63.** GATE ECE 1996

Open solution

Match the following.

**Group - I**

(a)Cascode amplifier
(b)Differential Amplifier
(c)Darlington pair common-collector Amplifier

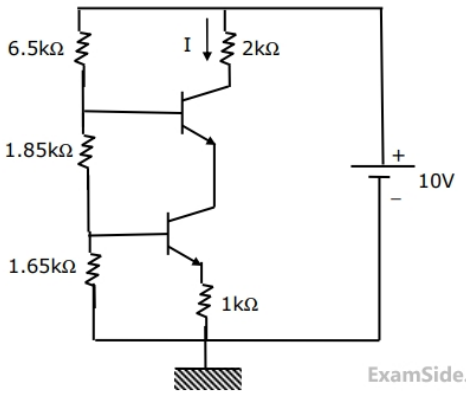
**Group - II**

(1)does not provide current gain
(2)is a wide band Amplifier
(3)has very low input impedance Emitter Amplifier and very high current gain
(4)has very high input impedance and very high current gain
(5)Provides high common mode voltage Rejection

**Q64.** GATE ECE 1992

Open solution

If the transistors in fig. has high value of β and a $v_{BE}$ of 0.65, the current I flowing through the 2 kilo ohms resistance will be _____.

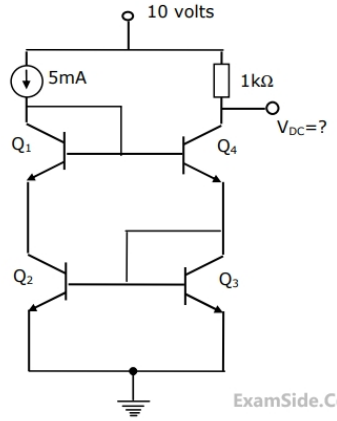



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**Q65.** GATE ECE 1991

In figure, all transistors are identical and have a high value of beta. The voltage VDC is equal to _____.

Open solution




**Q66.** GATE ECE 1990

Which of the following statements are correct for basic transistor Amplifier configurations?

(A) CB Amplifiers has low input impendence and low current gain.

(B) CC Amplifiers has low output impedence and a high current gain.

(C) CE Amplifier has very poor voltage gain but very high input impedance.

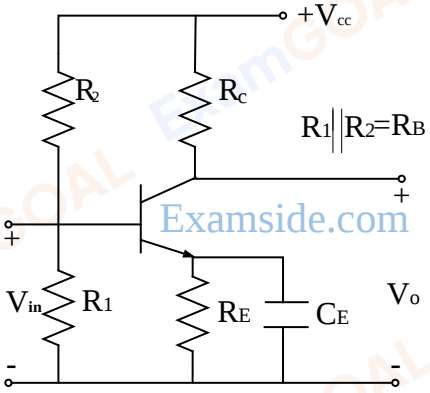
(D) The current gain of CB Amplifier is higher than the current gain of CC Amplifiers.

Open solution

**Q67.** GATE ECE 1990

For good stabilized biasing of the transistor of the CE Amplifier of fig, we should have

Open solution




(A) $\frac{R_E}{R_B} << 1$

(B) $\frac{R_E}{R_B} >> 1$

(C) $\frac{R_e}{R_B} << h_{FE}$

(D) $\frac{R_E}{R_B} >> h_{FE}$

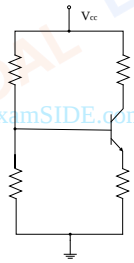
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**Q68.** GATE ECE 1989

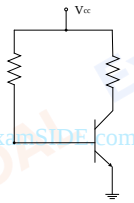
Of the four biasing circuits shown in Fig. For a BJT, indicate the one which can have maximum bias stability:

Open solution

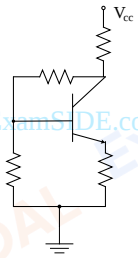
(A)



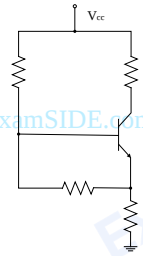

(B)




(C)




(D)




**Q69.** GATE ECE 1988

The quiescent collector current $I_C$, of a transistor is increased by changing resistance. As a result

Open solution

(A) $g_m$ will not be affected

(B) $g_m$ will decrease

(C) $g_m$ will increase

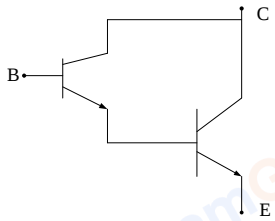
(D) $g_m$ will increase or decrease depending upon bias stability.

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**Q70.** GATE ECE 1988

Each transistor in the Darlington pair (see Fig. below) has $h_{fe}$ = 100. The overall $h_{fe}$ of the composite transistor neglecting the leakage current is

Open solution

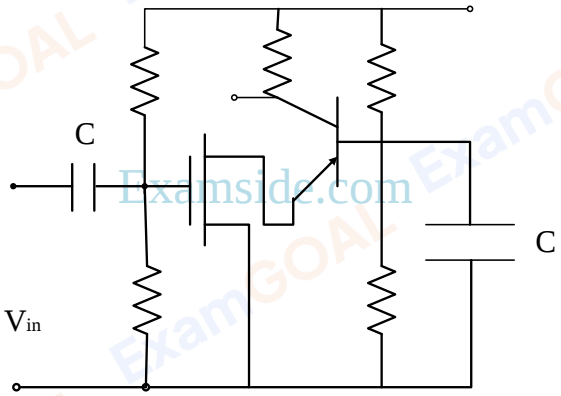



(A) 1000 (B) 10001 (C) 10100 (D) 10200

**Q71.** GATE ECE 1988

The amplifier circuit shown below uses a compostie transistor of a MOSFET and BIPOLAR in cascade. ALL capacitance are large. gm of the MOSFET = 2 mA/V , and $h_{fe}$ of the BIPOLAR = 99. The overall Transconductance $g_m$ of the composite transistor is

Open solution




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(A) 198 mA/V. (B) 9.9 mA/V. (C) 4.95 mA/V. (D) 1.98 mA/V.

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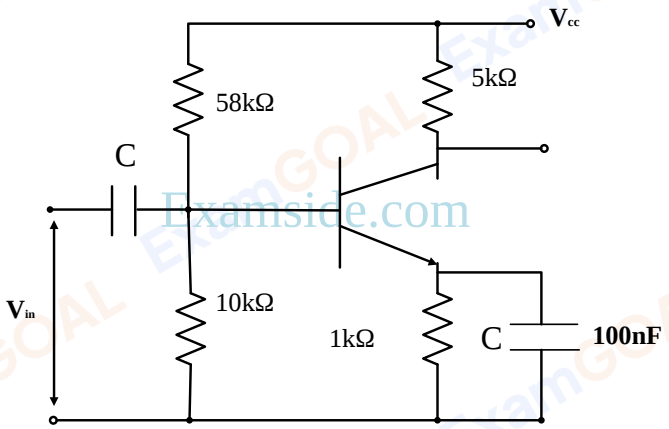
**Q72.** GATE ECE 1988

Open solution

The transistor in the amplifier shown below has following parameters:

$$h_{fe} = 100,\ h_{ie} = 2k\Omega,\ h_{re} = 0,$$

$h_{oe} = 0.05\ m\Omega$. $C$ is very large.
The output impedance is



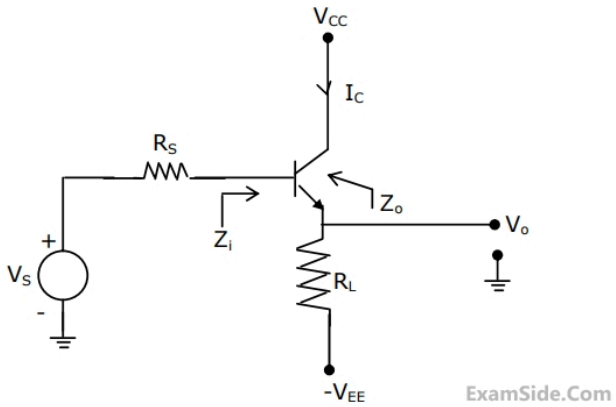

(A) $20k\Omega$. (B) $16k\Omega$. (C) $5k\Omega$. (D) $4k\Omega$.

## Marks 5 6 questions

**Q73.** GATE ECE 2001

Open solution

An emitter-follower amplifier is shown in the figure, $Z_i$ is the impedance looking into the base of the transistor and $Z_0$ is the impedance looking into the emitter of the transistor



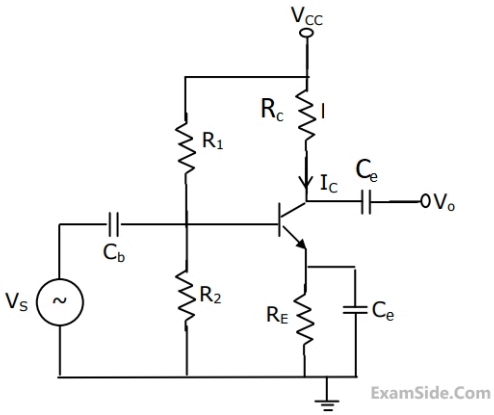

(a)Draw the small signal equivalent circuit of the amplifier.
(b)Obtain an expression for $z_i$.
(c)Obtain an expression for $z_0$.

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**Q74.** GATE ECE 2000

Open solution

For the amplifier of given figure,
$I_C = 1.3\,mA$, $R_C = 2\,k\Omega$, $R_E = 500\,\Omega$,
$V_T = 26\,mV$, $\beta = 100$, $V_{CC} = 15V$,
$V_s = 0.01\sin(\omega t)\,V$ and $C_b = C_C = 10\,\mu F$.

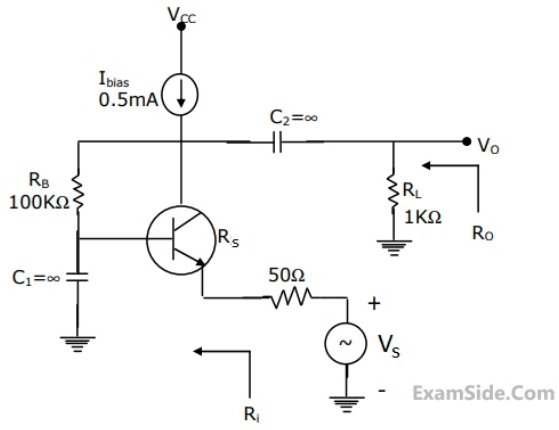



(a)What is the small-signal voltage gain, $A_V = V_0/V_s$?
(b)What is the approximate $A_{v,}\ if\ C_e$ is removed?
(c)What will $V_0\ be\ if\ C_b$ is short circuited?

**Q75.** GATE ECE 1999

Open solution

A bipolar junction transistor amplifier circuit is shown in the figure. Assume that the current source $I_{bias}$ is ideal, and the transistor has very large$\beta$, $r_b = 0$, and the $r_0 \to \infty$.



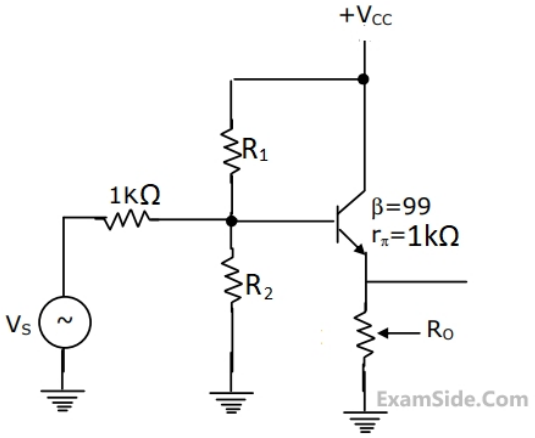

Determine the ac small-signal mid-band and voltage gain $(V_o/V_s)$, input resistance ($R_1$, and output resistance ($R_0$) of the circuit. Assume $V_T = 26\,mV$.

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**Q76.** GATE ECE 1998

Open solution

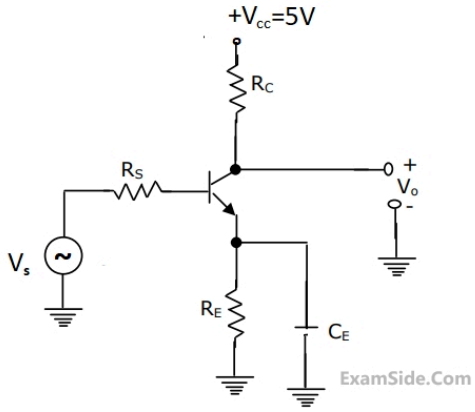
In the circuit of fig. Determine the resistance $R_o$ seen by the output terminals, ignore the effects of $R_1$ and $R_2$.




**Q77.** GATE ECE 1997

Open solution

The transistor in the circuit shown in the figure. is so biased (dc biasing N/W is not shown) that the dc collector current $I_C$ = 1mA. Supply is $V_{CC}$ = 5V.




The N/W components have following values, $R_C = 2k\Omega$,

$R_S = 1.4k\Omega$,

$R_E = 100\Omega$.

The transistor has specifications, $\beta = 100$

and base spreading resistance $r_{bb}{}^1 = 100\Omega$

$Assume\ \frac{KT}{q} = 25\,mV$

Evaluate input resistance $R_i$ for two cases. At a frequency of 10 kHz

(a)$C_E$, the bypass capacitor across $R_E$ is 25 $\mu F$

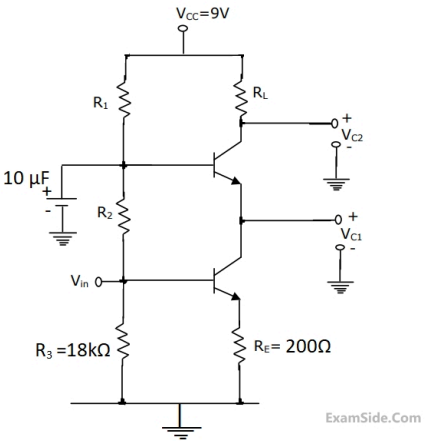
(b)The bypass capacitor $C_E$ is removed leaving $R_E$ unbypassed.

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**Q78.** GATE ECE 1997

In the cascade amplifier circuit shown below, determine the values of $R_1$, $R_2$ and $R_L$. Such that the quiescent current through the transistors is 1mA and the collector voltage $V_{c1}$ = 3V, and $V_{c2}$ = 6V. Tke $V_{BE}$ = 0.7V, Transistor $\beta$ to be hifgh and base currents to be negligible.

Open solution




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# Answer Key

## Marks 1

| | | | | |
|---|---|---|---|---|
| 1. **A** | 2. **C** | 3. **598.7** | 4. **0.5** | 5. **B** |
| 6. **34.7** | 7. **1** | 8. **C** | 9. **A** | 10. **B** |
| 11. **A** | 12. **D** | 13. **B** | 14. **A** | 15. **C** |
| 16. **C** | 17. **D** | 18. **C** | 19. **D** | 20. **C** |
| 21. **D** | 22. **A** | 23. **Compressed** | 24. **1** | 25. **C** |
| 26. **C** | | | | |

## Marks 2

| | | | | |
|---|---|---|---|---|
| 27. **A** | 28. **A** | 29. **A, C, D** | 30. *Scan QR* | 31. **800** |
| 32. **C** | 33. **50** | 34. **128** | 35. **0.9** | 36. **-1** |
| 37. **83.85** | 38. **5.785** | 39. **-237.76** | 40. **B** | 41. **0.5** |
| 42. **9.4** | 43. **B** | 44. **C** | 45. **D** | 46. **D** |
| 47. **C** | 48. **D** | 49. **A** | 50. **D** | 51. **A** |
| 52. **B** | 53. **C** | 54. **B** | 55. **A** | 56. **C** |
| 57. **B** | 58. **D** | 59. **A** | 60. **C** | 61. **D** |
| 62. **D** | 63. **(a) - (2), (b) - (5), (c) - (4)** | 64. **1** | 65. **5** | 66. **A, B** |
| 67. **B** | 68. **A** | 69. **C** | 70. **D** | 71. **D** |
| 72. **D** | | | | |

## Marks 5

| | | | | |
|---|---|---|---|---|
| 73. **(b) $z_i$ =** $\frac{1}{\frac{1}{R_B}+\frac{1}{h_{ie}+(1+h_{ie})R_E}}$ **(c)** $Z_0 = \frac{1}{\frac{1}{R_E}+\left(\frac{1+h_{fe}}{h_{ie}}\right)}$ | 74. **(a)** $A_v = -101$ **(b)** $A_v = -4$ **(c)** $V_o = -1.01\ \sin(\omega_t)V$ | 75. $\frac{V_o}{V_s} = \frac{g_m}{\left(\frac{1}{R_B}+\frac{1}{R_L}\right)\left(\frac{1+50}{r_\pi}+50\ g_m\right)}$ $R_i = \frac{1+\frac{50}{r_\pi}+50\ g_m}{\frac{1}{r_\pi}+g_m}$ $R_o = R_B$ | 76. **$R_0$=20Ω** | 77. **(a)$R_i$ =** $\frac{14K+j200k}{1+j50}$ **(b)$R_i$ =** $14K\Omega$ |
| 78. $R_1 = 106\,K\Omega$ $R_2 = 56\,K\Omega$ $R_L = 3\,K\Omega$ | | | | |

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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